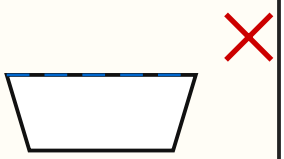
Robotics Project #13 · Templates 5/5 (1:1 scale)

Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

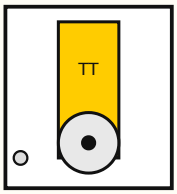
1 Cut and fold the barrow tray (sheet 1). Glue the tabs inside.



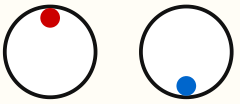
2 Skewer axle through a straw under the tray; glue a wheel on each end. Wheels roll freely.



3 Hot-glue a hip plate (sheet 3) on each flat side of the TT motor, motor upright, shaft in the Ø30 hole.



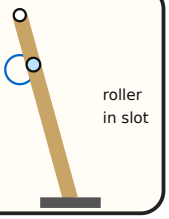
4 Glue crank A on one shaft end, crank B on the other: the pins must point opposite ways.



5 Glue a 22 mm hip peg in each hip hole. Add a 5 mm straw spacer, then the leg (hip sleeve in its Ø5.5 hole).



6 Crank pin + straw roller sits in the leg slot. Glue button stoppers on peg and pin ends.



7 Glue torso covers, arms at the shoulders, hands onto the handles with both feet on the table.

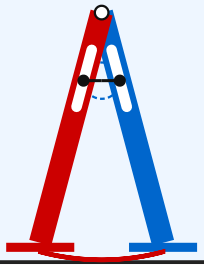


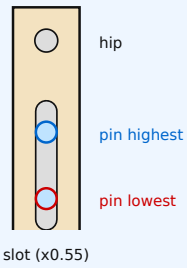
8 2xAA + switch under the tray, wires along a handle to the motor. Switch on and he walks!



ON = walk!

How the legs walk





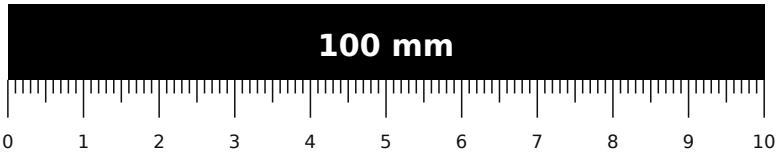
The motor turns both cranks, half a turn apart. The crank pin (with a straw roller) slides up and down in the leg slot and pushes the leg front and back. The leg swings like a pendulum from the hip peg: the foot (red) sweeps 56 mm along the table and rises only 4 mm. One foot pushes back while the other swings forward - and his hands push the barrow! Roller never leaves the slot.

Parts checklist

- ☐ 3 mm cardboard
- ☐ 3 ice-cream sticks
- ☐ 2 wheels (caps/65 mm)
- ☐ Skewers 3 mm
- ☐ Paper straw
- ☐ TT motor (double shaft)
- ☐ 2xAA holder + batteries
- ☐ Slide switch + wire
- ☐ Rubber bands (feet)
- ☐ Ping-pong ball + cap
- ☐ Hot glue (grown-up)
- ☐ Pencil, scissors

Safety first!

- A grown-up uses the hot glue gun, craft knife, and pokes the holes.
- Snip the sharp skewer tips off.
- Fingers away from the cranks when ON.
- Batteries the right way round; never join the wires; take them out after.
- Keep small parts from little kids.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.